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Russo

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(54) **ASSEMBLIES COMPRISING MEMORY CELLS AND SELECT GATES; AND METHODS OF FORMING ASSEMBLIES**

(71) Applicant: **Micron Technology, Inc.**, Boise, ID (US)

(72) Inventor: **Ugo Russo**, Boise, ID (US)

(73) Assignee: **Micron Technology, Inc.**, Boise, ID (US)

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(58) **Field of Classification Search**
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See application file for complete search history.

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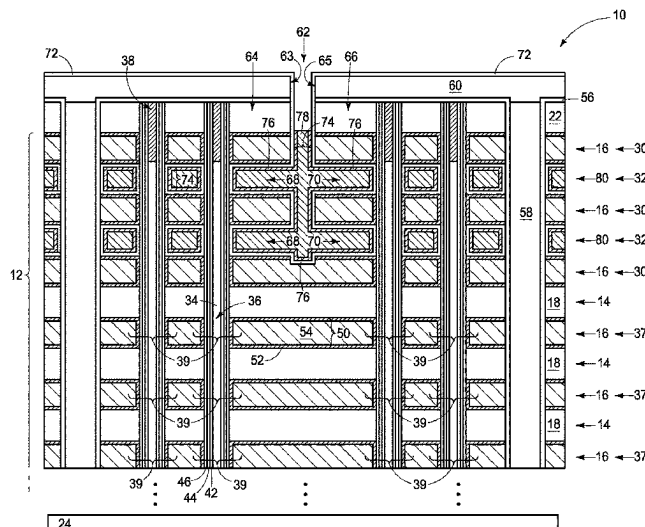
Primary Examiner — Alfredo Bermudez Lozada

(74) *Attorney, Agent, or Firm* — Wells St. John P.S.

(57) **ABSTRACT**

Some embodiments include an assembly having a stack of alternating dielectric levels and conductive levels. Channel material pillars extend through the stack. Some of the channel material pillars are associated with a first sub-block, and others of the channel material pillars are associated with a second sub-block. Memory cells are along the channel material pillars. An insulative level is over the stack. A select gate configuration is over the insulative level. The select gate configuration includes a first conductive gate structure associated with the first sub-block, and includes a second conductive gate structure associated with the second sub-block. The first and second conductive gate structures are laterally spaced from one another by an intervening insulative region. The first and second conductive gate structures have vertically-spaced conductive regions, and have vertically-extending conductive structures which electrically couple the vertically-spaced conductive regions to one another. Some embodiments include methods of forming assemblies.

28 Claims, 17 Drawing Sheets



Related U.S. Application Data

- division of application No. 16/810,058, filed on Mar. 5, 2020, now Pat. No. 11,170,826, which is a division of application No. 15/926,427, filed on Mar. 20, 2018, now Pat. No. 10,614,862.
- (60) Provisional application No. 62/609,884, filed on Dec. 22, 2017.
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H10B 41/35 (2023.01)
H10B 43/27 (2023.01)
G11C 11/56 (2006.01)

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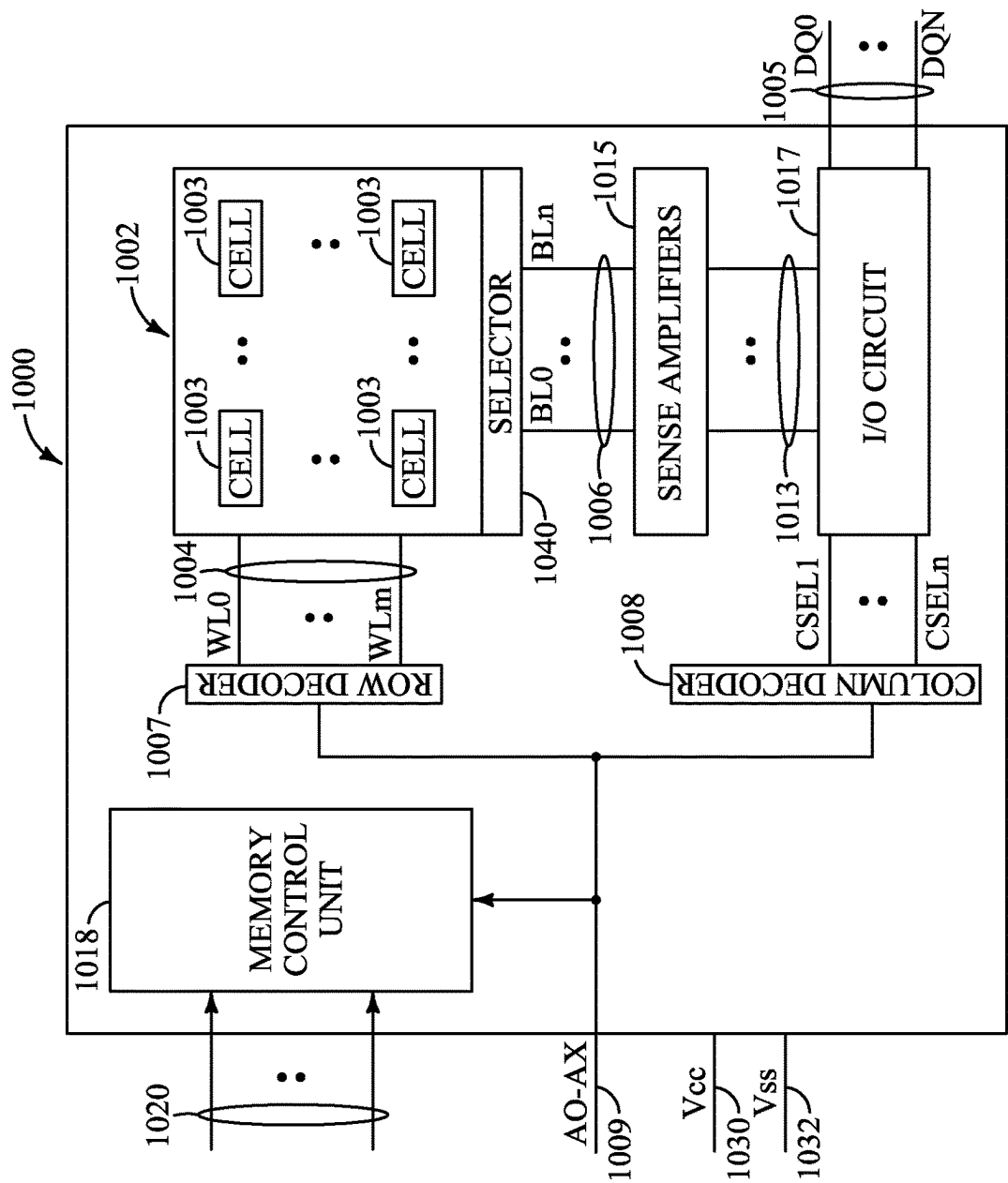
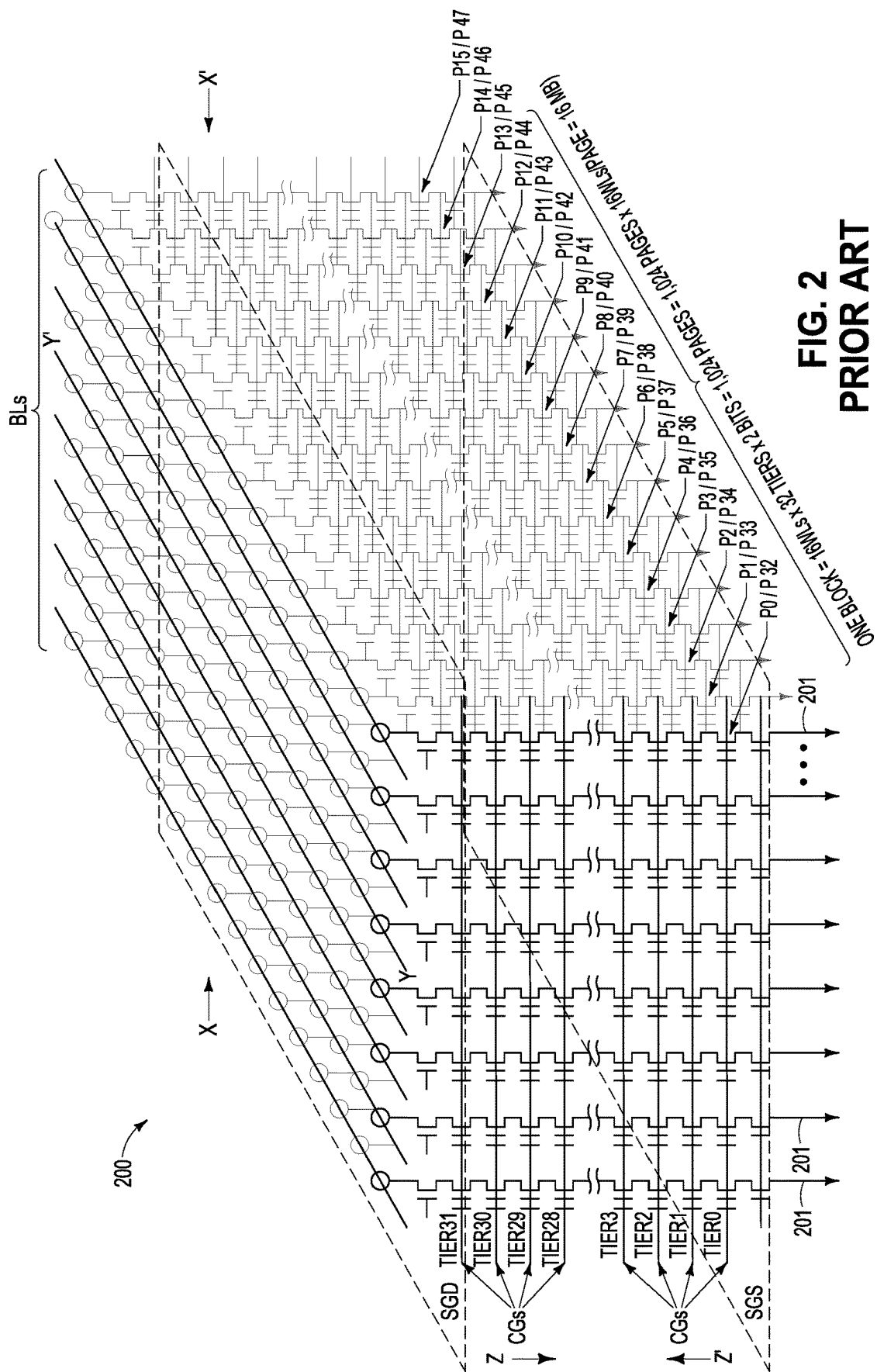
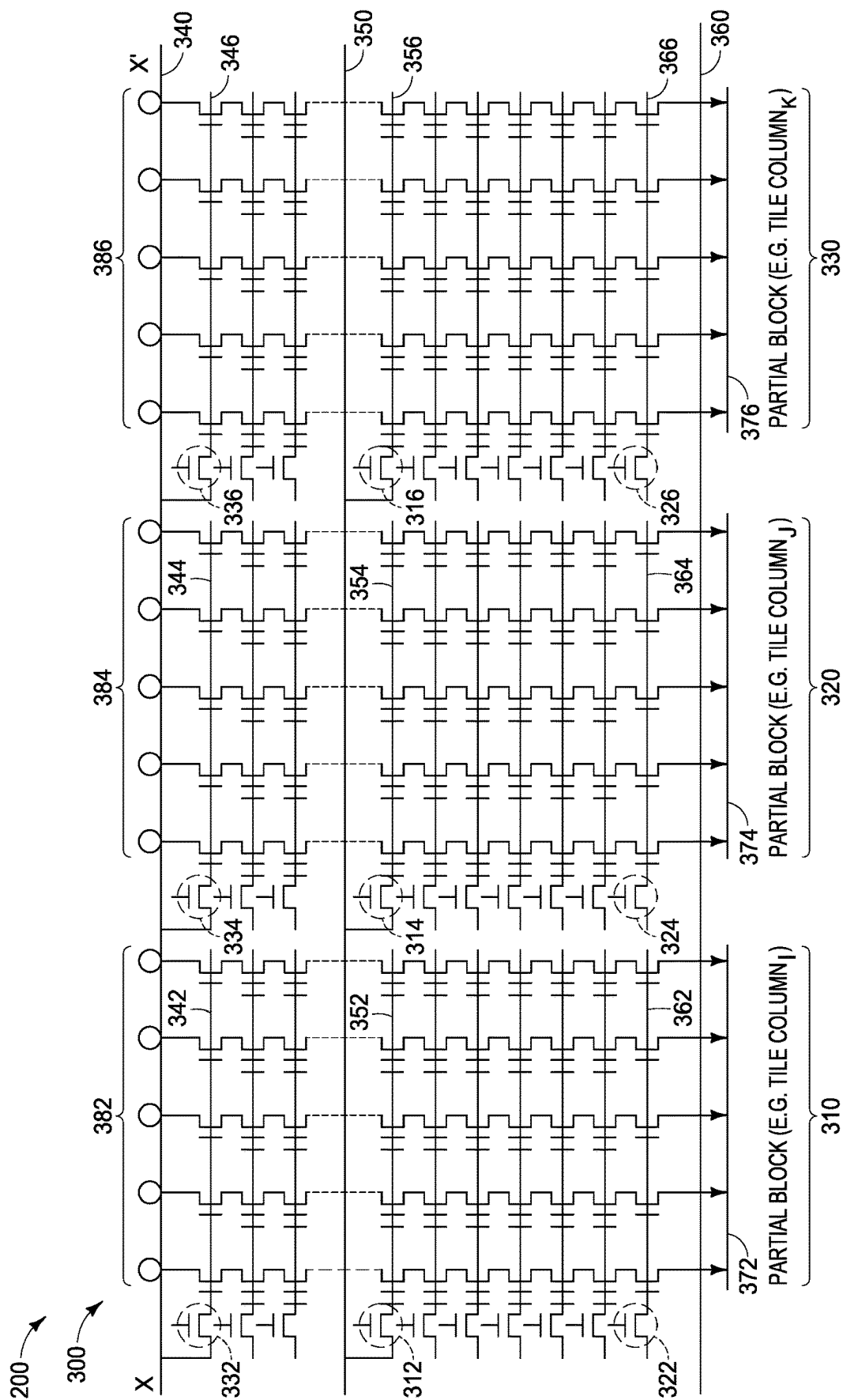


FIG. 1
PRIOR ART





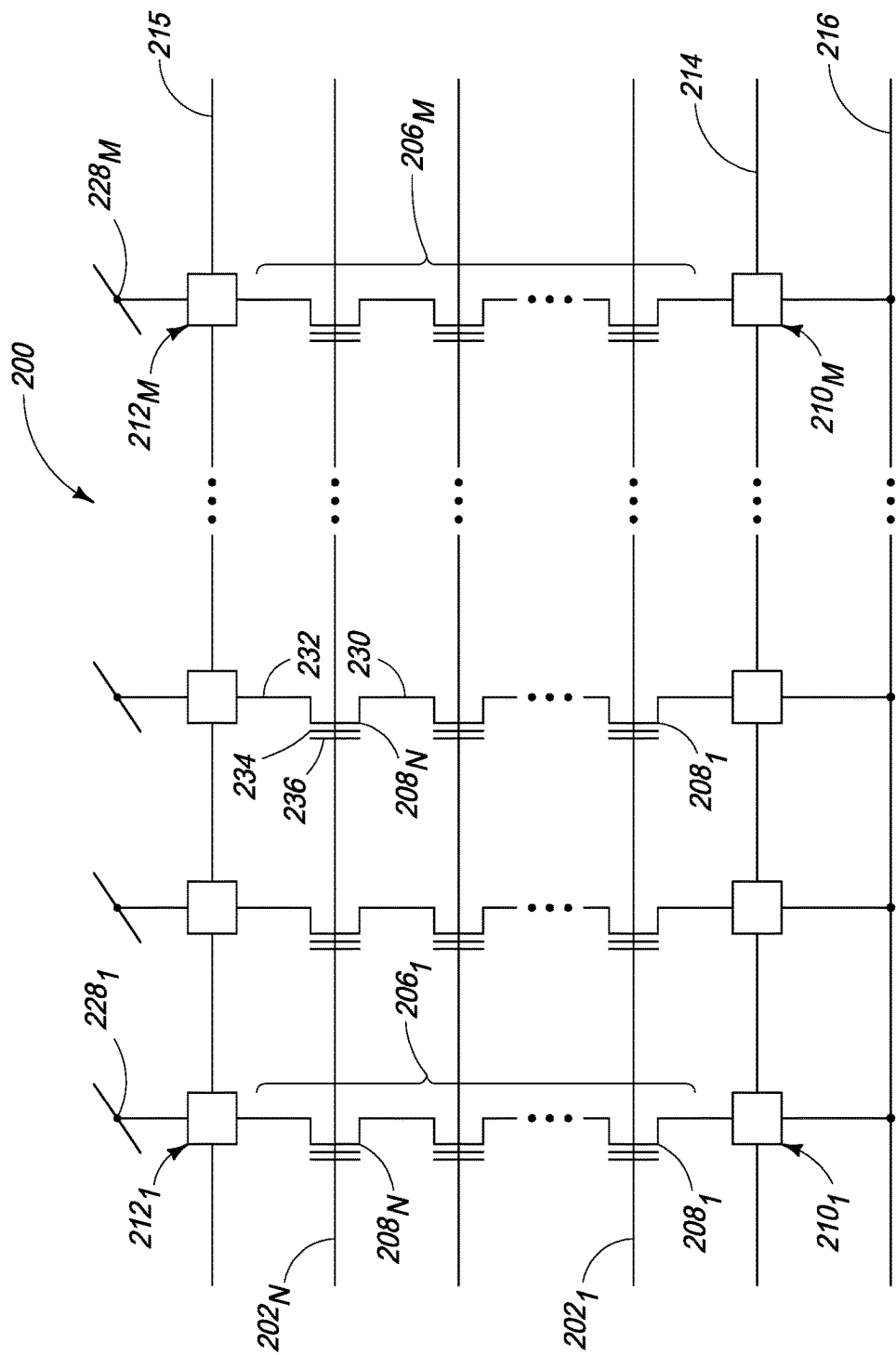


FIG. 4
PRIOR ART

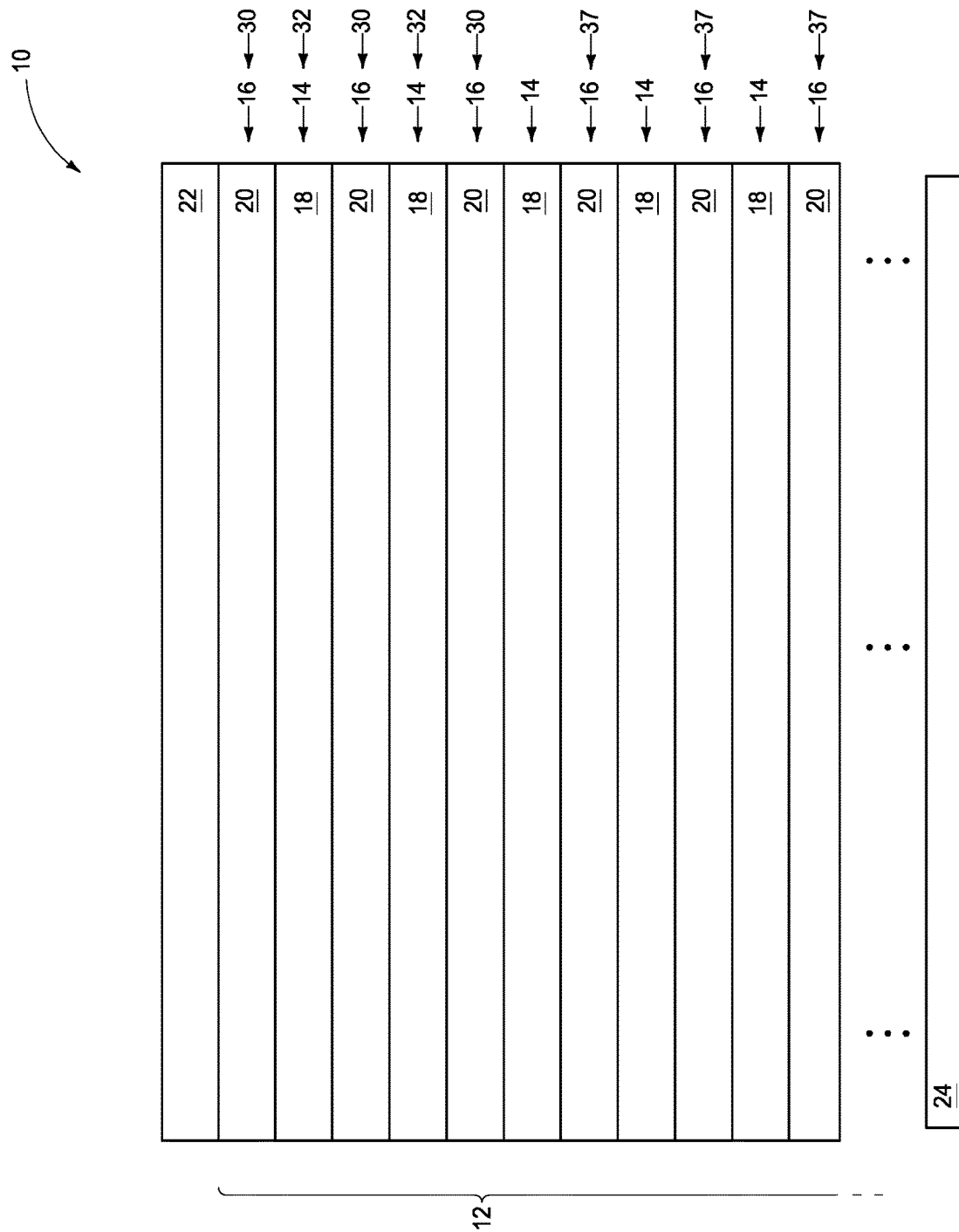
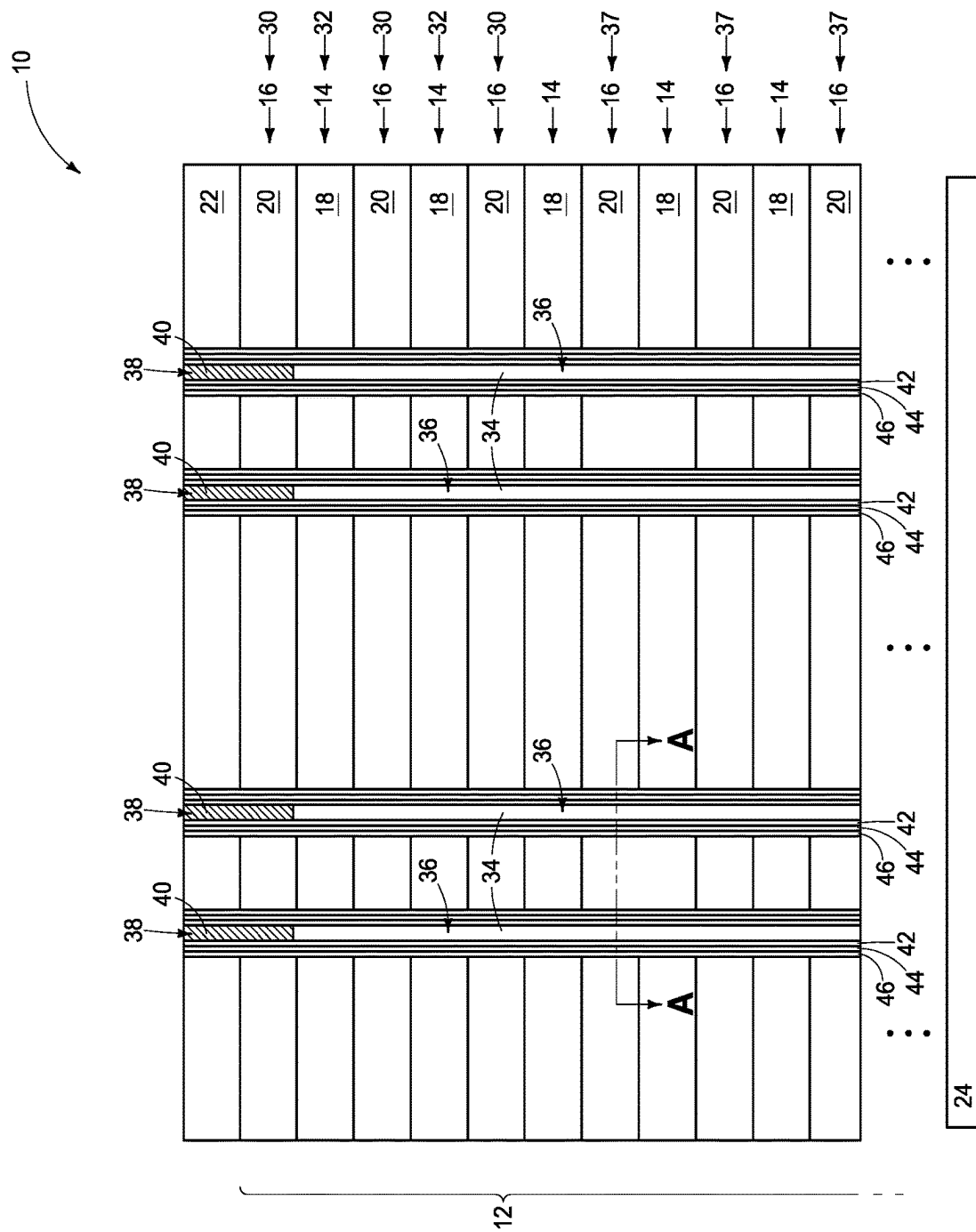


FIG. 5



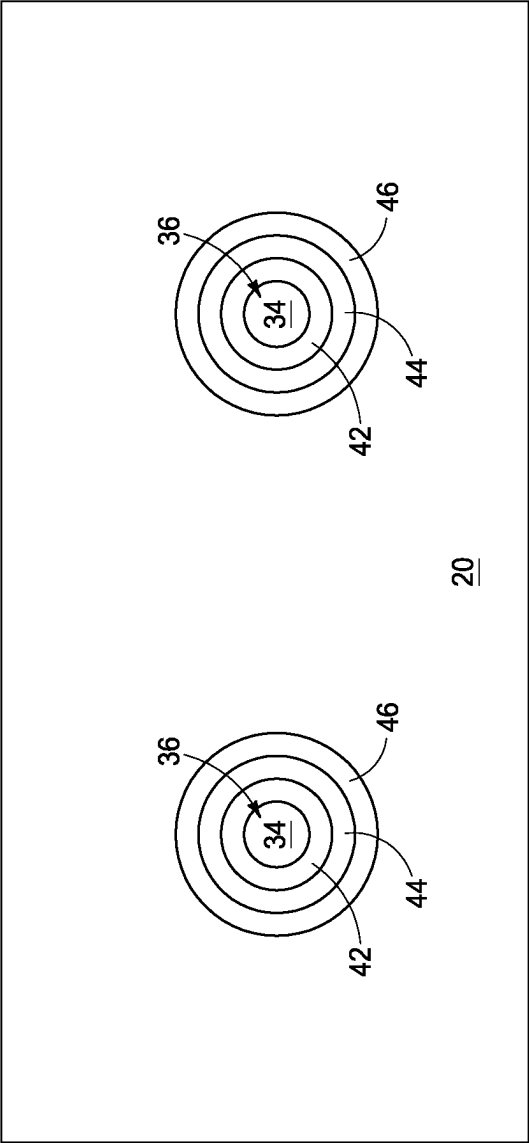


FIG. 6A

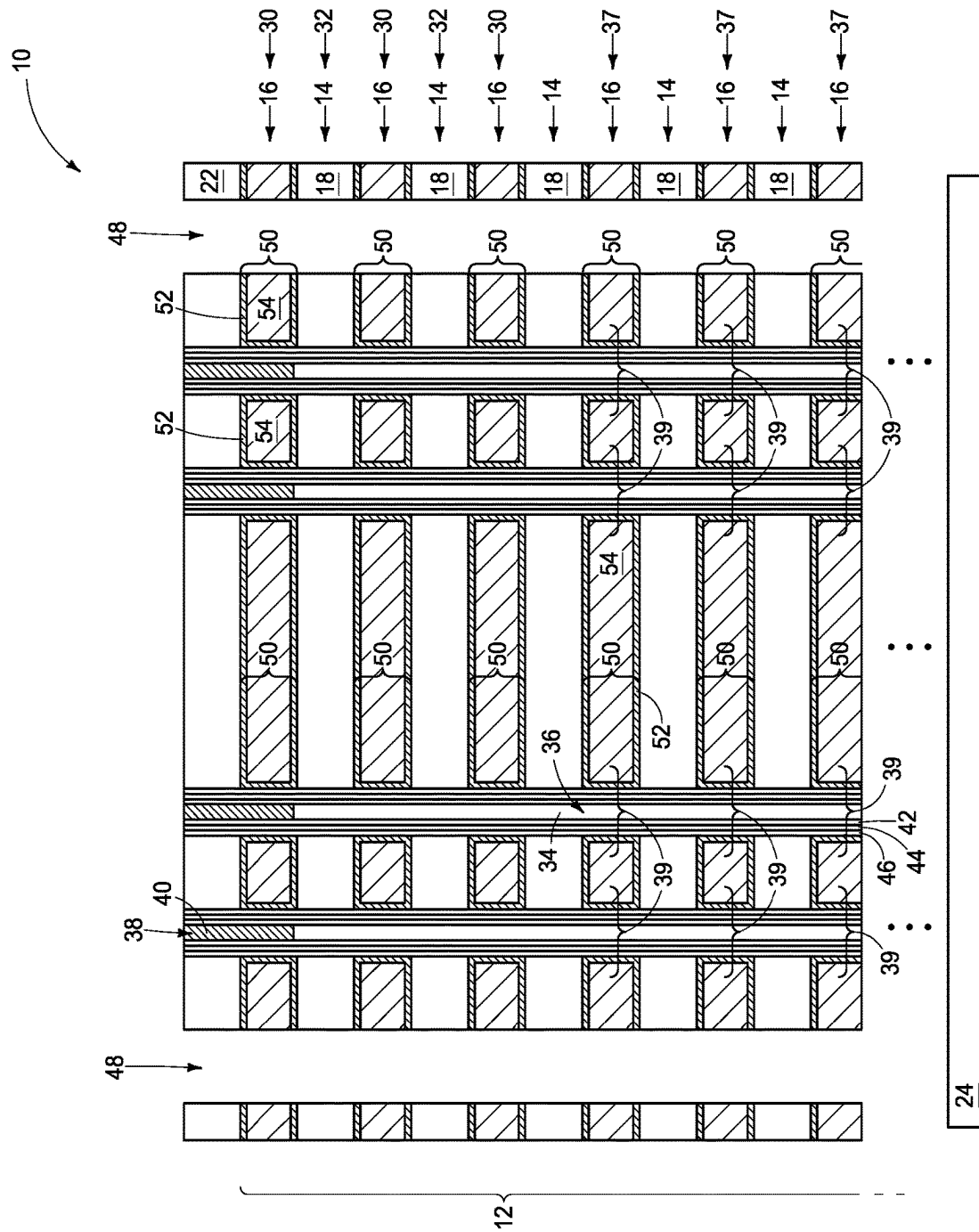


FIG. 7

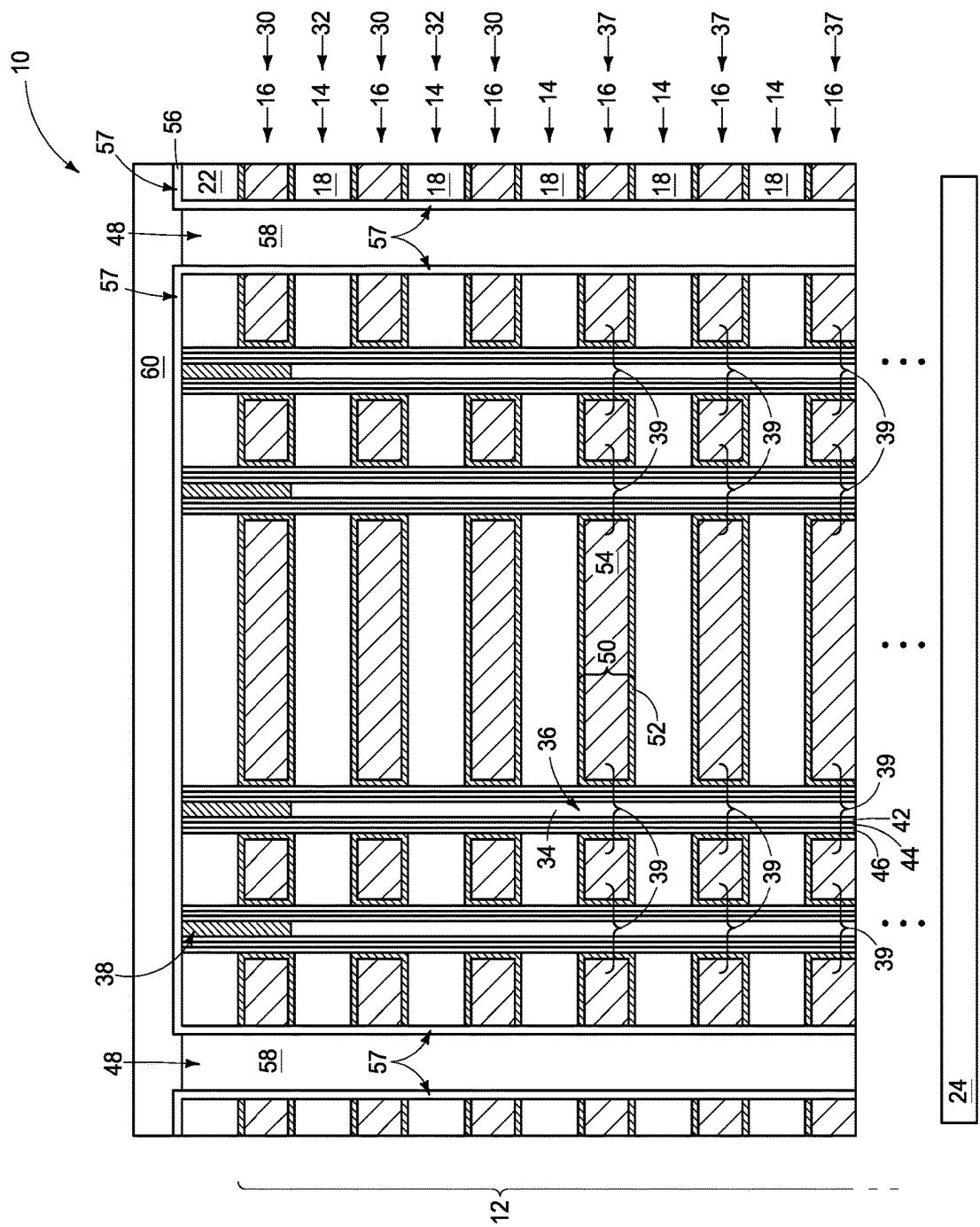


FIG. 8

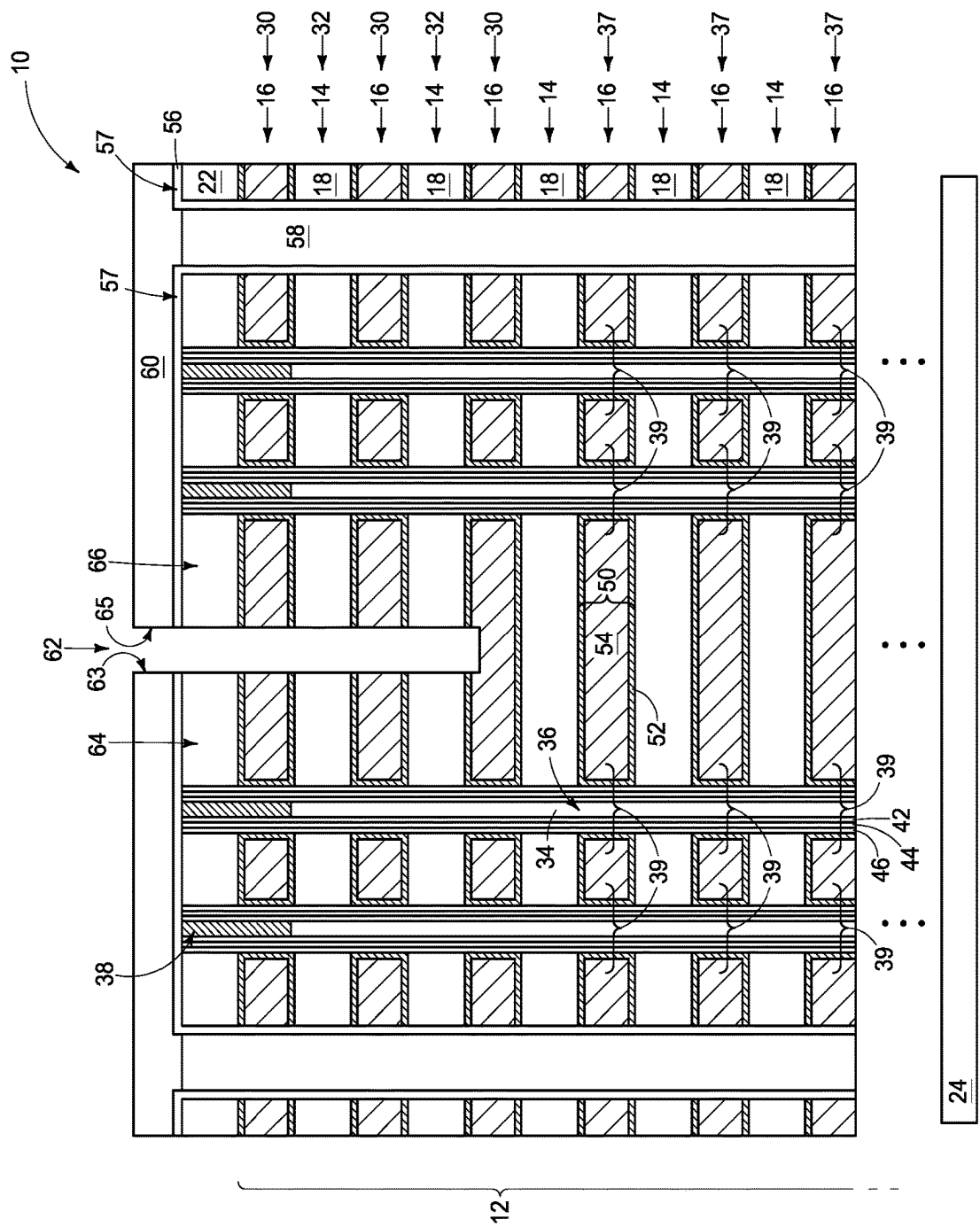


FIG. 9

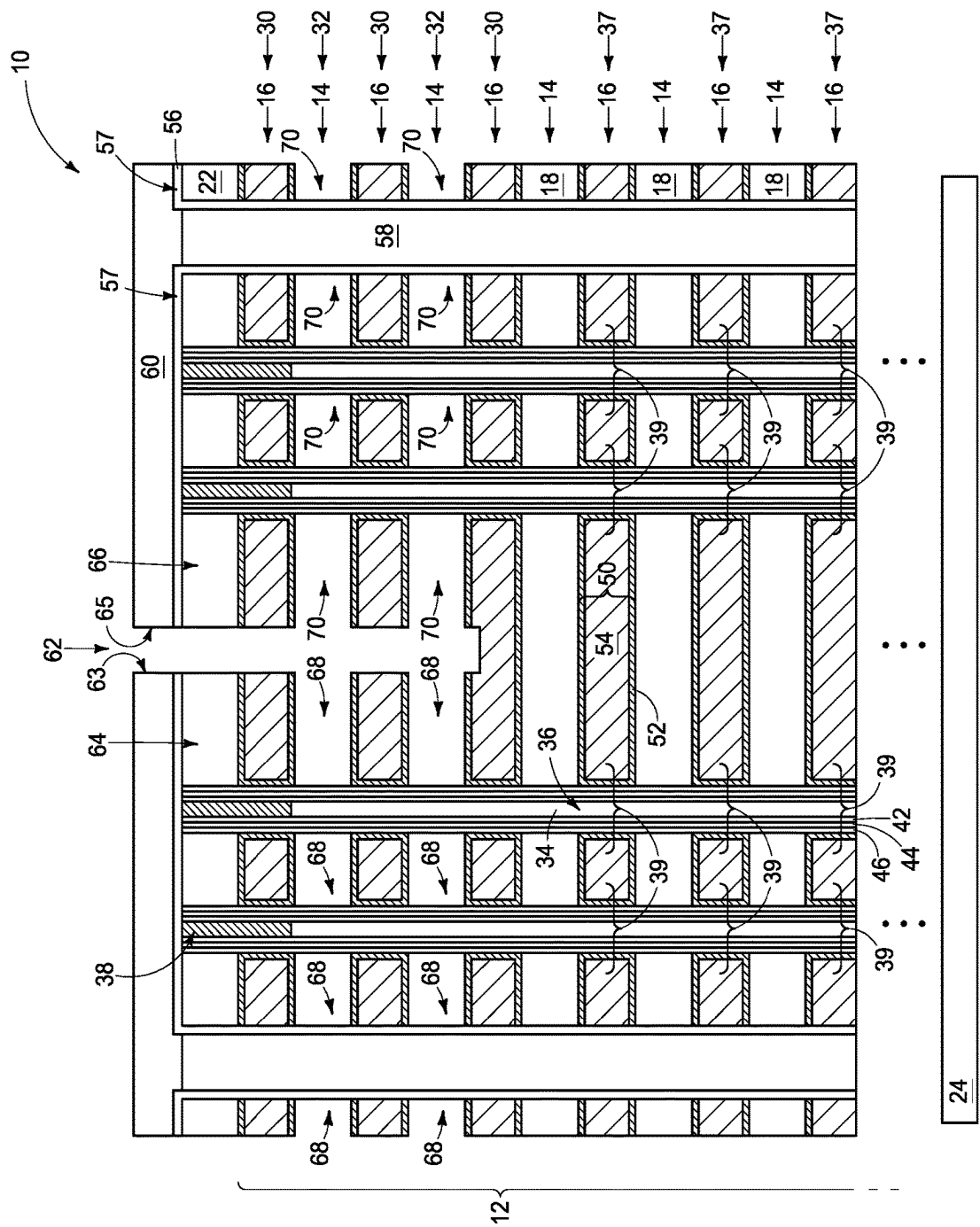


FIG. 10

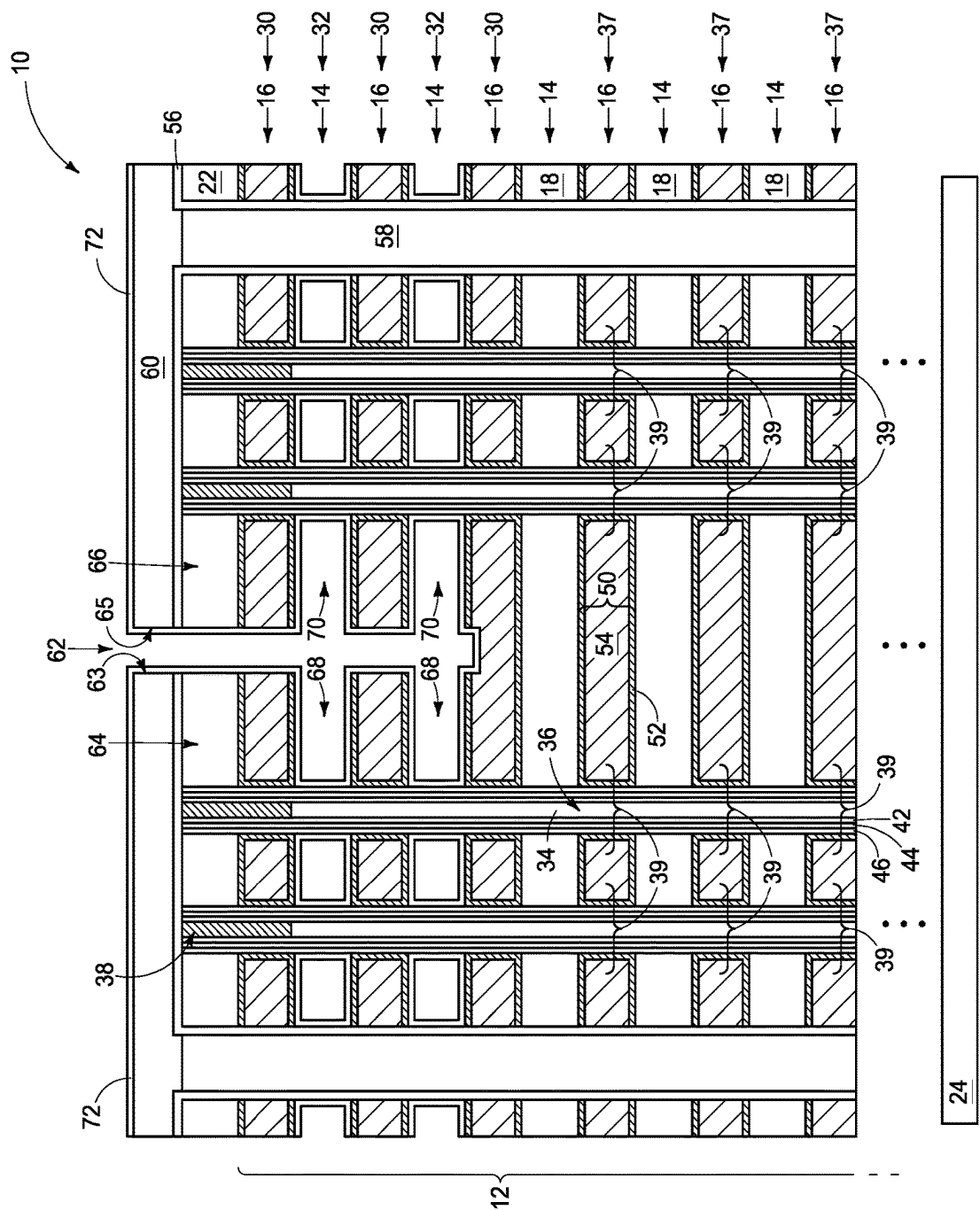


FIG. 11

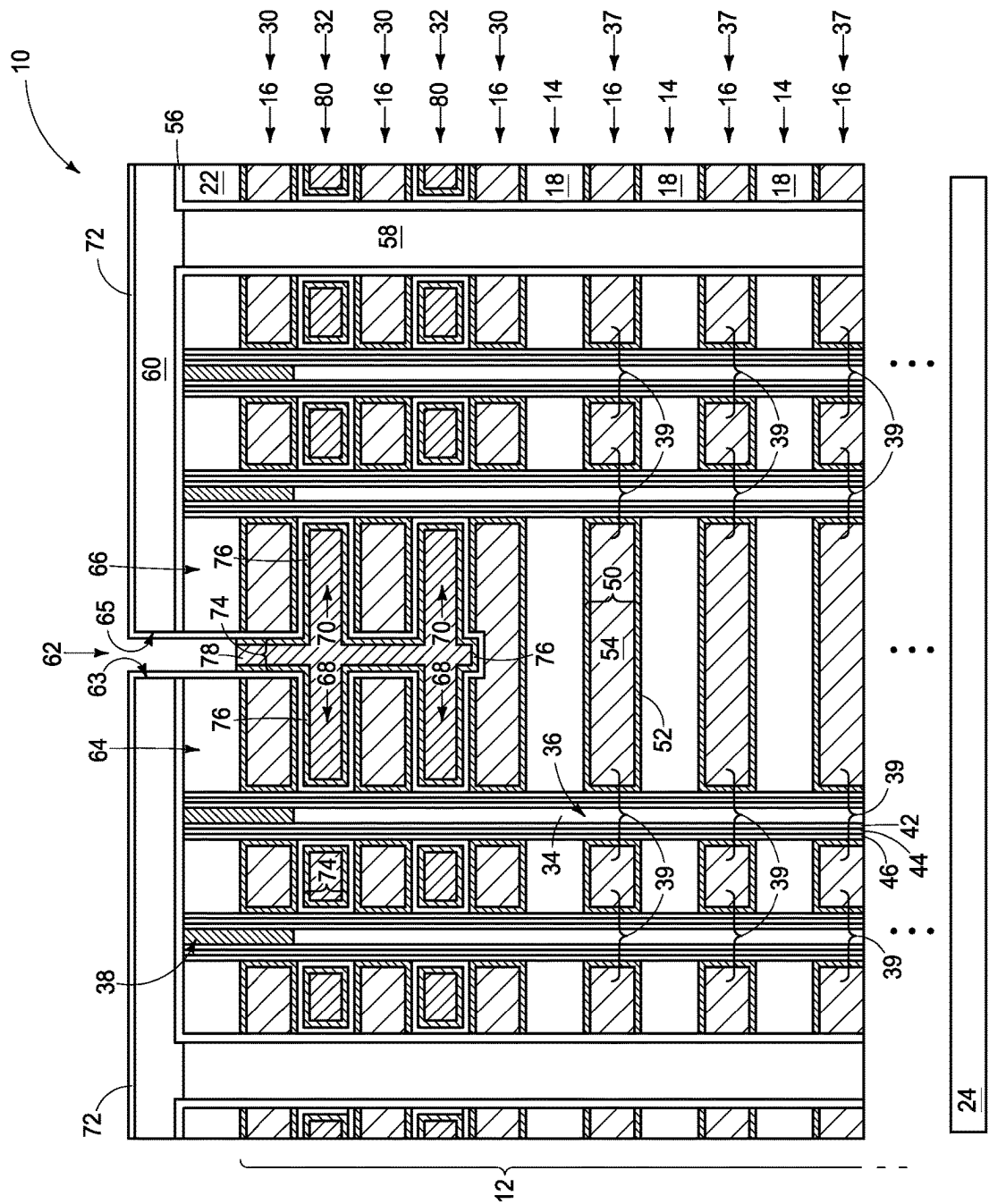


FIG. 12

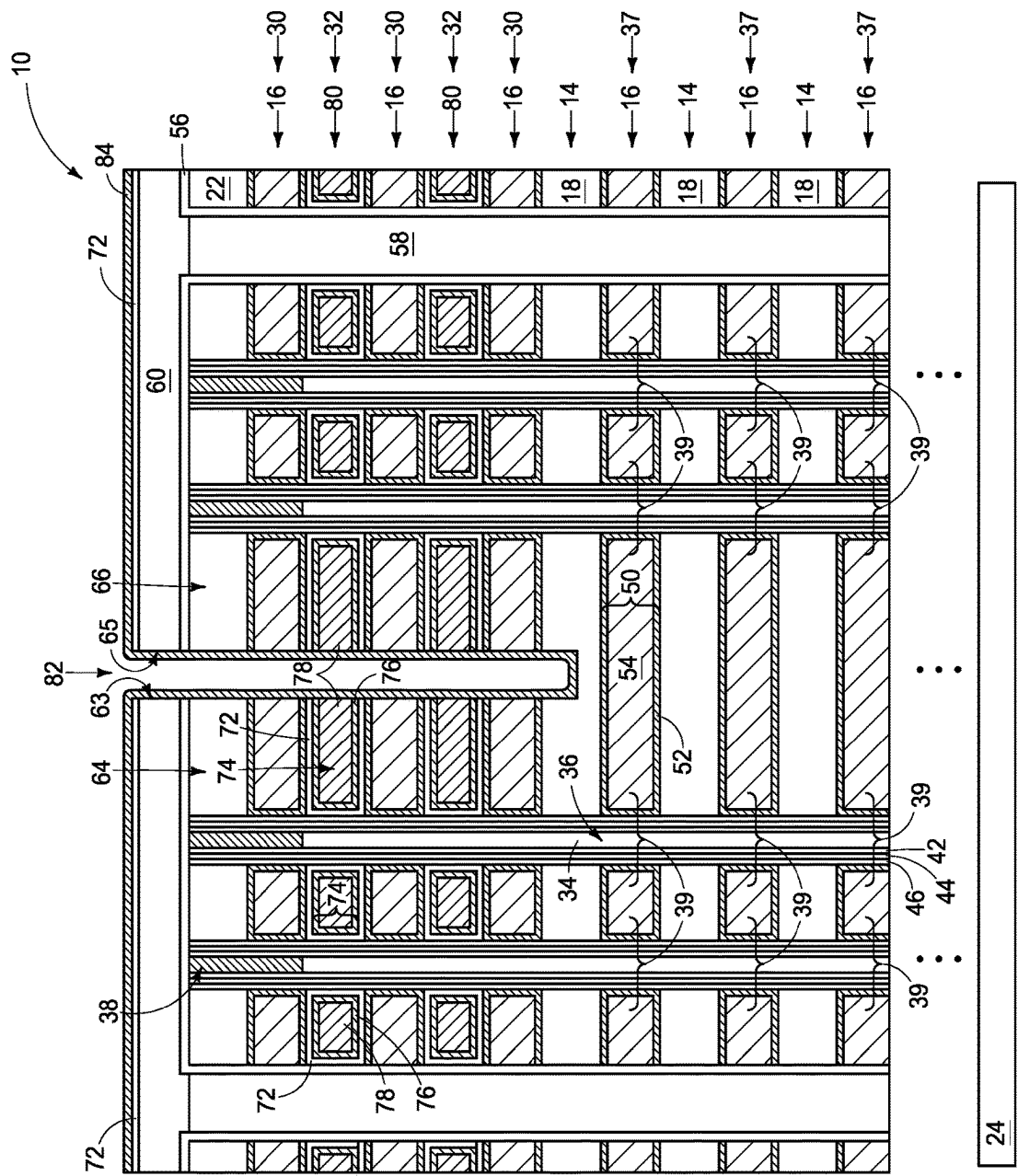


FIG. 14

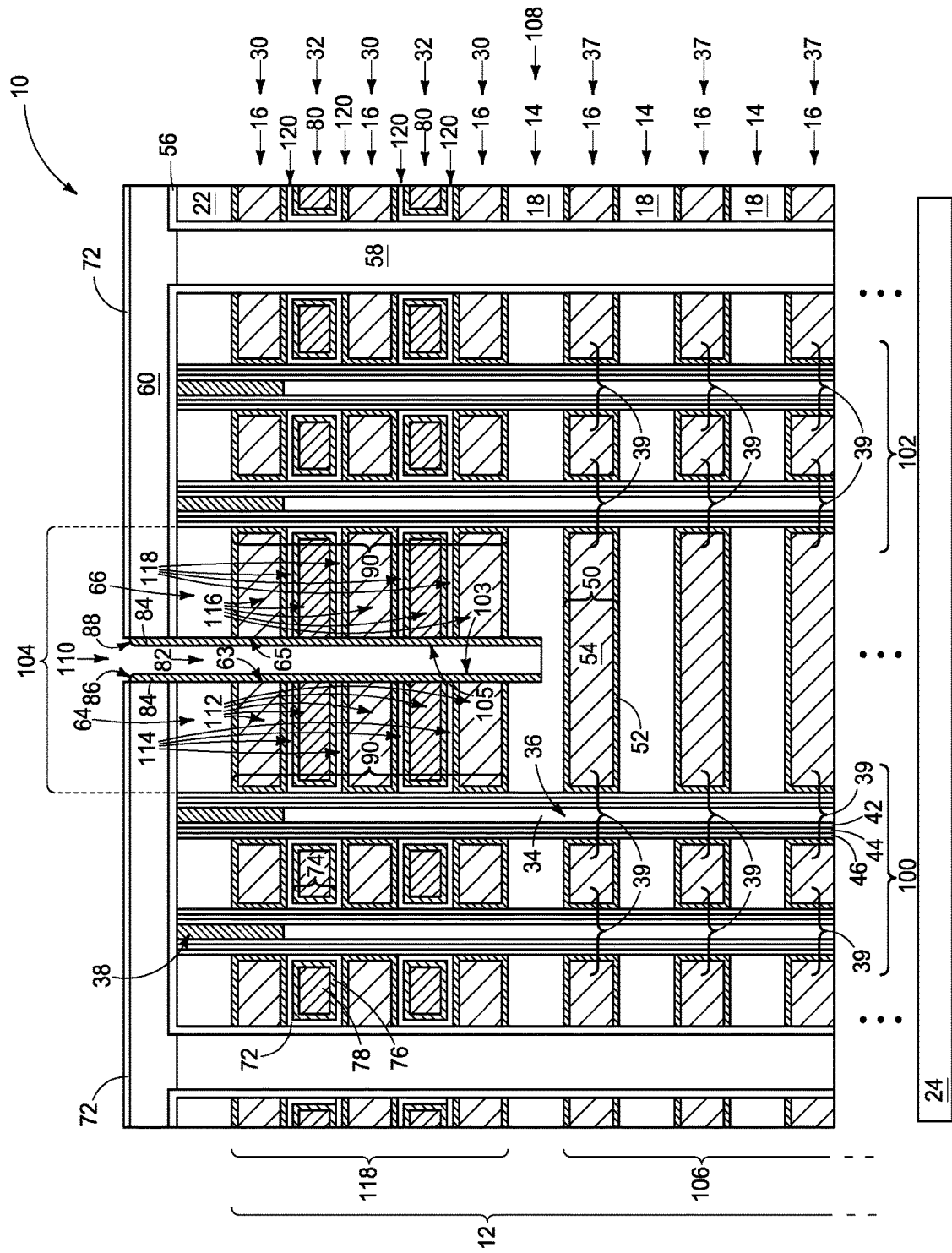


FIG. 15

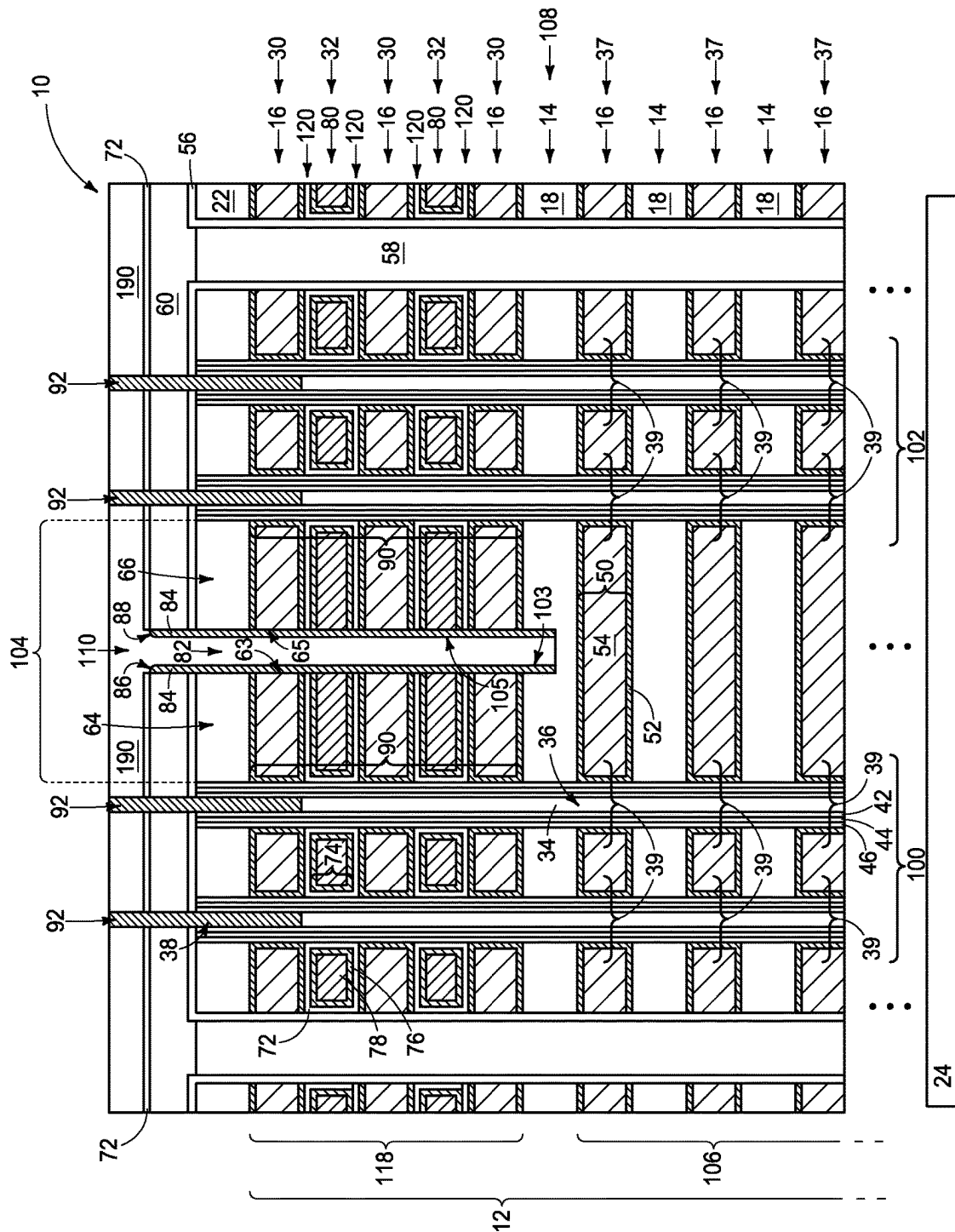


FIG. 16

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ASSEMBLIES COMPRISING MEMORY CELLS AND SELECT GATES; AND METHODS OF FORMING ASSEMBLIES

RELATED PATENT DATA

This patent resulted from a continuation of U.S. patent application Ser. No. 17/499,218, filed Oct. 12, 2021, which resulted from a divisional of U.S. patent application Ser. No. 16/810,058, filed Mar. 5, 2020, now U.S. Pat. No. 11,170,826, which resulted from a divisional of U.S. patent application Ser. No. 15/926,427, filed Mar. 20, 2018, now U.S. Pat. No. 10,614,862, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/609,884, filed Dec. 22, 2017, the disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD

Assemblies comprising memory cells and select gates; such as assemblies comprising drain-side select gates (SGDs) over NAND memory cells. Methods of forming integrated assemblies.

BACKGROUND

Memory provides data storage for electronic systems. Flash memory is one type of memory, and has numerous uses in modern computers and devices. For instance, modern personal computers may have BIOS stored on a flash memory chip. As another example, it is becoming increasingly common for computers and other devices to utilize flash memory in solid state drives to replace conventional hard drives. As yet another example, flash memory is popular in wireless electronic devices because it enables manufacturers to support new communication protocols as they become standardized, and to provide the ability to remotely upgrade the devices for enhanced features.

NAND may be a basic architecture of flash memory, and may be configured to comprise vertically-stacked memory cells.

Before describing NAND specifically, it may be helpful to more generally describe the relationship of a memory array within an integrated arrangement. FIG. 1 shows a block diagram of a prior art device **1000** which includes a memory array **1002** having a plurality of memory cells **1003** arranged in rows and columns along with access lines **1004** (e.g., wordlines to conduct signals WL₀ through WLM) and first data lines **1006** (e.g., bitlines to conduct signals BL₀ through BLN). Access lines **1004** and first data lines **1006** may be used to transfer information to and from the memory cells **1003**. A row decoder **1007** and a column decoder **1008** decode address signals A₀ through A_X on address lines **1009** to determine which ones of the memory cells **1003** are to be accessed. A sense amplifier circuit **1015** operates to determine the values of information read from the memory cells **1003**. An I/O circuit **1017** transfers values of information between the memory array **1002** and input/output (I/O) lines **1005**. Signals DQ₀ through DQ_N on the I/O lines **1005** can represent values of information read from or to be written into the memory cells **1003**. Other devices can communicate with the device **1000** through the I/O lines **1005**, the address lines **1009**, or the control lines **1020**. Memory control unit **1018** controls memory operations performed on the memory cells **1003** utilizing signals on the control lines **1020**. The device **1000** can receive supply voltage signals V_{cc} and V_{ss} on a first supply line **1030** and a second supply line **1032**,

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respectively. The device **1000** includes a select circuit **1040** and an input/output (I/O) circuit **1017**. The select circuit **1040** can respond, via the I/O circuit **1017**, to signals CSEL₁ through CSEL_N to select signals on the first data lines **1006** and the second data lines **1013** that can represent the values of information to be read from or to be programmed into the memory cells **1003**. The column decoder **1008** can selectively activate the CSEL₁ through CSEL_N signals based on the A₀ through A_X address signals on the address lines **1009**. The select circuit **1040** can select the signals on the first data lines **1006** and the second data lines **1013** to provide communication between the memory array **1002** and the I/O circuit **1017** during read and programming operations.

The memory array **1002** of FIG. 1 may be a NAND memory array, and FIG. 2 shows a block diagram of a three-dimensional NAND memory device **200** which may be utilized for the memory array **1002** of FIG. 1. The device **200** comprises a plurality of strings of charge-storage devices. In a first direction (Z-Z'), each string of charge-storage devices may comprise, for example, thirty-two charge-storage devices stacked over one another with each charge-storage device corresponding to one of, for example, thirty-two tiers (e.g., Tier₀-Tier₃₁). The charge-storage devices of a respective string may share a common channel region, such as one formed in a respective pillar of semiconductor material (e.g., polysilicon) about which the string of charge-storage devices is formed. In a second direction (X-X'), each first group of, for example, sixteen first groups of the plurality of strings may comprise, for example, eight strings sharing a plurality (e.g., thirty-two) of access lines (i.e., "global control gate (CG) lines", also known as wordlines, WLs). Each of the access lines may couple the charge-storage devices within a tier. The charge-storage devices coupled by the same access line (and thus corresponding to the same tier) may be logically grouped into, for example, two pages, such as P₀/P₃₂, P₁/P₃₃, P₂/P₃₄ and so on, when each charge-storage device comprises a cell capable of storing two bits of information. In a third direction (Y-Y'), each second group of, for example, eight second groups of the plurality of strings, may comprise sixteen strings coupled by a corresponding one of eight data lines. The size of a memory block may comprise 1,024 pages and total about 16 MB (e.g., 16 WLs×32 tiers×2 bits=1,024 pages/block, block size=1,024 pages×16 KB/page=16 MB). The number of the strings, tiers, access lines, data lines, first groups, second groups and/or pages may be greater or smaller than those shown in FIG. 2.

FIG. 3 shows a cross-sectional view of a memory block **300** of the 3D NAND memory device **200** of FIG. 2 in an X-X' direction, including fifteen strings of charge-storage devices in one of the sixteen first groups of strings described with respect to FIG. 2. The plurality of strings of the memory block **300** may be grouped into a plurality of subsets **310**, **320**, **330** (e.g., tile columns), such as tile column₁, tile column₂ and tile column_K, with each subset (e.g., tile column) comprising a "partial block" of the memory block **300**. A global drain-side select gate (SGD) line **340** may be coupled to the SGDs of the plurality of strings. For example, the global SGD line **340** may be coupled to a plurality (e.g., three) of sub-SGD lines **342**, **344**, **346** with each sub-SGD line corresponding to a respective subset (e.g., tile column), via a corresponding one of a plurality (e.g., three) of sub-SGD drivers **332**, **334**, **336**. Each of the sub-SGD drivers **332**, **334**, **336** may concurrently couple or cut off the SGDs of the strings of a corresponding partial block (e.g., tile column) independently

of those of other partial blocks. A global source-side select gate (SGS) line **360** may be coupled to the SGSs of the plurality of strings. For example, the global SGS line **360** may be coupled to a plurality of sub-SGS lines **362**, **364**, **366** with each sub-SGS line corresponding to the respective subset (e.g., tile column), via a corresponding one of a plurality of sub-SGS drivers **322**, **324**, **326**. Each of the sub-SGS drivers **322**, **324**, **326** may concurrently couple or cut off the SGSs of the strings of a corresponding partial block (e.g., tile column) independently of those of other partial blocks. A global access line (e.g., a global CG line) **350** may couple the charge-storage devices corresponding to the respective tier of each of the plurality of strings. Each global CG line (e.g., the global CG line **350**) may be coupled to a plurality of sub-access lines (e.g., sub-CG lines) **352**, **354**, **356** via a corresponding one of a plurality of sub-string drivers **312**, **314** and **316**. Each of the sub-string drivers may concurrently couple or cut off the charge-storage devices corresponding to the respective partial block and/or tier independently of those of other partial blocks and/or other tiers. The charge-storage devices corresponding to the respective subset (e.g., partial block) and the respective tier may comprise a “partial tier” (e.g., a single “tile”) of charge-storage devices. The strings corresponding to the respective subset (e.g., partial block) may be coupled to a corresponding one of sub-sources **372**, **374** and **376** (e.g., “tile source”) with each sub-source being coupled to a respective power source.

The NAND memory device **200** is alternatively described with reference to a schematic illustration of FIG. 4.

The memory array **200** includes wordlines **2021** to **202x**, and bitlines **2281** to **228M**.

The memory array **200** also includes NAND strings **2061** to **206M**. Each NAND string includes charge-storage transistors **2081** to **208x**. The charge-storage transistors may use floating gate material (e.g., polysilicon) to store charge, or may use charge-trapping material (such as, for example, silicon nitride, metallic nanodots, etc.) to store charge.

The charge-storage transistors **208** are located at intersections of wordlines **202** and strings **206**. The charge-storage transistors **208** represent non-volatile memory cells for storage of data. The charge-storage transistors **208** of each NAND string **206** are connected in series source-to-drain between a source-select device (e.g., source-side select gate, SGS) **210** and a drain-select device (e.g., drain-side select gate, SGD) **212**. Each source-select device **210** is located at an intersection of a string **206** and a source-select line **214**, while each drain-select device **212** is located at an intersection of a string **206** and a drain-select line **215**. The select devices **210** and **212** may be any suitable access devices, and are generically illustrated with boxes in FIG. 1.

A source of each source-select device **210** is connected to a common source line **216**. The drain of each source-select device **210** is connected to the source of the first charge-storage transistor **208** of the corresponding NAND string **206**. For example, the drain of source-select device **2101** is connected to the source of charge-storage transistor **2081** of the corresponding NAND string **2061**. The source-select devices **210** are connected to source-select line **214**.

The drain of each drain-select device **212** is connected to a bitline (i.e., digit line) **228** at a drain contact. For example, the drain of drain-select device **2121** is connected to the bitline **2281**. The source of each drain-select device **212** is connected to the drain of the last charge-storage transistor **208** of the corresponding NAND string **206**. For example,

the source of drain-select device **2121** is connected to the drain of charge-storage transistor **208x** of the corresponding NAND string **2061**.

The charge-storage transistors **208** include a source **230**, a drain **232**, a charge-storage region **234**, and a control gate **236**. The charge-storage transistors **208** have their control gates **236** coupled to a wordline **202**. A column of the charge-storage transistors **208** are those transistors within a NAND string **206** coupled to a given bitline **228**. A row of the charge-storage transistors **208** are those transistors commonly coupled to a given wordline **202**.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a block diagram of a prior art memory device having a memory array with memory cells.

FIG. 2 shows a schematic diagram of the prior art memory array of FIG. 1 in the form of a 3D NAND memory device.

FIG. 3 shows a cross sectional view of the prior art 3D NAND memory device of FIG. 2 in an X-X' direction.

FIG. 4 is a schematic of a prior art NAND memory array.

FIGS. 5-16 are diagrammatic cross-sectional views of an example construction at example process stages of an example method for forming an example assembly.

FIG. 6A is a cross-sectional view along the line A-A of FIG. 6.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

Some embodiments include new methods of fabricating drain-side select gates (SGDs), and new architectures formed utilizing such methods. An example method is described with reference to FIGS. 5-16, and an example architecture is described with reference to FIG. 16.

Referring to FIG. 5, a construction **10** includes a stack **12** of alternating first and second levels **14** and **16**. The first levels **14** comprise dielectric material **18** (e.g., silicon dioxide), and the second levels **16** comprise sacrificial material **20** (e.g., silicon nitride). The dielectric material **18** may be referred to as first dielectric material to distinguish it from other dielectric materials that will be formed at later process stages.

The levels **14** and **16** may be of any suitable thicknesses; and may be the same thickness as one another, or different thicknesses relative to one another. In some embodiments, the levels **14** and **16** may have vertical thicknesses within a range of from about 10 nanometers (nm) to about 300 nm.

A masking material **22** is over the stack **12**. The masking material may comprise any suitable composition(s), and in some embodiments may comprise amorphous carbon. The masking material **22** may be utilized to form a hard mask.

Two or more upper second levels **16** of the stack **12** may be referred to as gate levels **30**, in that they are ultimately incorporated into drain-side select gates (SGDs). Others of the second levels **16** may be referred to as wordline levels **37**, in that they are ultimately incorporated into wordlines. The wordlines may comprise control gates for the memory cells; such as vertically-stacked memory cells of NAND strings. The number of memory cell levels in the individual strings is determined by the number of wordlines. The NAND strings may comprise any suitable number of memory cell levels. For instance, the NAND strings may have 8 memory cell levels, 16 memory cell levels, 32 memory cell levels, 64 memory cell levels, 512 memory cell levels, 1024 memory cell levels, etc.

The first levels **14** between the gate levels **30** may be referred to as intervening levels **32** (or as intervening first levels).

The stack **12** shown to be supported over a base **24**. The base **24** may comprise semiconductor material; and may, for example, comprise, consist essentially of, or consist of monocrystalline silicon. The base **24** may be referred to as a semiconductor substrate. The term “semiconductor substrate” means any construction comprising semiconductive material, including, but not limited to, bulk semiconductive materials such as a semiconductive wafer (either alone or in assemblies comprising other materials), and semiconductive material layers (either alone or in assemblies comprising other materials). The term “substrate” refers to any supporting structure, including, but not limited to, the semiconductor substrates described above. In some applications, the base **24** may correspond to a semiconductor substrate containing one or more materials associated with integrated circuit fabrication. Such materials may include, for example, one or more of refractory metal materials, barrier materials, diffusion materials, insulator materials, etc.

A gap is provided between the stack **12** and the base **24** to indicate that other components and materials may be provided between the stack **12** and the base **24**. Such other components and materials may comprise additional levels of the stack, a source line level, source-side select gates (SGSs), etc.

Referring to FIG. 6, channel material **34** is provided in the form of pillars **36**. In the illustrated embodiment, the channel material pillars **36** are solid pillars. In other embodiments, the channel material pillars may be so-called hollow pillars (i.e., may comprise channel material surrounding an insulative core).

The channel material pillars **36** are capped by conductive interconnects **38**. The conductive interconnects **38** comprise electrically conductive material **40**. The material **40** may comprise any suitable electrically conductive composition (s), such as, for example, one or more of various metals (e.g., titanium, tungsten, cobalt, nickel, platinum, ruthenium, etc.), metal-containing compositions (e.g., metal silicide, metal nitride, metal carbide, etc.), and/or conductively-doped semiconductor materials (e.g., conductively-doped silicon, conductively-doped germanium, etc.). In some embodiments, the material **40** may comprise, consist of, or consist of conductively-doped silicon.

The channel material **34** may comprise any suitable appropriately-doped semiconductor material(s); and in some embodiments may comprise one or more of silicon, germanium, III/V semiconductor materials (e.g., gallium phosphide), semiconductor oxides, etc.

The channel material pillars **34** extend into the stack **12**, and in the shown embodiment extend through two of the gate levels **30**, and two of the intervening levels **32**. Generally, the channel material pillars **34** extend through at least one of the gate levels **30**, and through at least one of the intervening levels **32**.

Tunneling material **42**, charge-storage material **44**, and charge-blocking material **46** are formed along the channel material pillars **36**.

The tunneling material **16** is sometimes referred to as gate dielectric. The tunneling material **16** may comprise any suitable composition(s); and in some embodiments may comprise, for example, one or more of silicon dioxide, aluminum oxide, hafnium oxide, zirconium oxide, etc.

The charge-storage material **18** may comprise any suitable composition(s); and in some embodiments may comprise charge-trapping materials, such as silicon nitride, sili-

con oxynitride, conductive nanodots, etc. In alternative embodiments (not shown), charge-storage material **26** may be configured as floating gate material (such as, for example, polycrystalline silicon).

The charge-blocking material **46** may comprise any suitable composition(s); and in some embodiments may comprise one or more of silicon dioxide, aluminum oxide, hafnium oxide, zirconium oxide, etc.

FIG. 6A shows a cross-section along the line A-A of FIG. 6, and shows an example configuration of the channel material pillars **36**, the tunneling material **42**, the charge-storage material **44** and the charge-blocking material **46**.

Referring to FIG. 7, slits **48** are formed to extend through the stack **12**. The slits **48** may be configured as trenches extending in and out of the page relative to the cross-sectional view of FIG. 7. After the slits **48** are formed, the sacrificial material **20** (FIG. 6) is removed and replaced with conductive material **50**.

The conductive material **50** may comprise any suitable electrically conductive composition(s), such as, for example, one or more of various metals (e.g., titanium, tungsten, cobalt, nickel, platinum, ruthenium, etc.), metal-containing compositions (e.g., metal silicide, metal nitride, metal carbide, etc.), and/or conductively-doped semiconductor materials (e.g., conductively-doped silicon, conductively-doped germanium, etc.). In the shown embodiment, the conductive material **50** comprises a first composition **52** and a second composition **54**. The first composition **52** may comprise, consist essentially, or consist of a metal nitride (e.g., titanium nitride); and the second composition **54** may comprise, consist essentially of, or consist of a metal (e.g., tungsten). In some embodiments, the conductive material **50** may be referred to as a first conductive material to distinguish it from other conductive materials formed at later process stages.

The conductive material **50** forms wordlines along the wordline levels **37**. Regions of the wordlines along the channel material pillars **36** become control gates of memory cells **39**; with such memory cells comprising the channel material **34**, tunneling material **42**, charge-storage material **44** and charge-blocking material **46**. The memory cells may be vertically-stacked NAND memory cells of a three-dimensional NAND memory array. The NAND memory cells **39** along a common channel material pillar **36** may be a string of the NAND memory cells.

In some embodiments, barrier material (not shown) may be provided between the conductive material **50** and the charge-blocking material **46**. The barrier material may be along regions where the conductive material **50** is adjacent the charge-blocking material **46**. The charge-blocking material **46** may provide a mechanism to block charge from migrating from the charge-trapping material **44** to the control gates of memory cells **39**, and the barrier material may provide a mechanism which prevents back-migration of electrons from the control gates into the charge-trapping material. The barrier material may be formed as liners extending along outer edges of the conductive material **50** in some embodiments. The barrier material may comprise any suitable composition(s); and in some embodiments may comprise one or more of aluminum oxide, hafnium oxide, zirconium oxide, niobium oxide, etc.

Referring to FIG. 8, the slits **48** are filled with insulative materials **56** and **58**. In some embodiments, the material **56** may be referred to as outer liner **57** provided along edges of the slits **48**, and the material **58** may be referred to as an inner material provided within the slits along the outer liner **57**.

The insulative material **56** may comprise any suitable composition or combination of compositions; and in some embodiments may comprise, consist essentially of, or consist of silicon nitride. The insulative material **58** may comprise any suitable composition or combination of compositions; and in some embodiments may comprise, consist essentially of, or consist of silicon dioxide. Accordingly, in some embodiments the insulative **58** may comprise a same composition as the insulative material **18** of the dielectric levels **14**.

In the illustrated embodiment, a protective capping material **60** is formed across upper surfaces of the insulative material **58**. The protective capping material **60** may protect the insulative material **58** from being etched during a subsequent etch utilized to remove insulative material **18** (with the subsequent etch being shown and described relative to FIG. **10**). If the insulative material **58** comprises a different composition relative to the material **18**, and would not be removed with the subsequent etched utilized to remove the material **18**, then the protective capping layer **60** may be omitted.

Referring to FIG. **9**, an opening **62** is formed to extend through at least some of the gate levels **30**, and through the intervening levels **32**. In the illustrated embodiment, the opening **62** extends only partially into the bottommost of the gate levels. In other embodiments, the opening **62** may extend entirely through such bottommost of the gate levels.

The opening **62** may be referred to as a first opening distinguish it from another opening formed at a later process stage.

The first opening **62** may be considered to comprise a first side **63** and an opposing second side **65** along the cross-section of FIG. **9**. The opening **62** may be considered to define a first gate **64** along the first side **63** of the opening, and to define a second gate **66** along the second side **65** of the opening. The first and second gates **64** and **66** defined by the opening **62** will ultimately become SGD gates. However, such gates are not fully fabricated at the process stage of FIG. **9**. Instead, the locations of the gates **64** and **66** are defined by the formation of the opening **62**.

The first opening **62** may have any suitable shape. In some embodiments, the opening may be a trench which extends in and out of the page relative to the cross-section of FIG. **9**.

Referring to FIG. **10**, the dielectric material **18** is removed from the intervening levels **32** to form first voids **68** along the first side **63** of the opening **62**, and to form second voids **70** along the second side **65** of the opening **62**. In some embodiments, the insulative material **18** may comprise silicon dioxide (as discussed above). In such embodiments, the voids **68** and **70** may be formed with a wet etch of the silicon dioxide.

The protective capping material **60** may protect the insulative material **58** from being removed during the etching of material **18**, as discussed above in explaining the purpose of the protective capping material **60**.

Referring to FIG. **11**, the voids **68** and **70** are lined with dielectric material **72**. In some embodiments, the dielectric materials **18** and **72** may be referred to as first and second dielectric materials, respectively, to distinguish them from one another.

The dielectric material **72** may comprise any suitable composition(s). For instance, in some embodiments the dielectric material **72** may comprise, consist essentially of, or consist of silicon dioxide. As another example, in some embodiments the dielectric material **72** may comprise a high-k oxide, where the term "high-k" means a dielectric constant greater than that of silicon oxide. For instance, in

some embodiments the dielectric material **72** may comprise, consist essentially of, or consist of one or more of aluminum oxide, hafnium oxide, zirconium oxide, etc.

In some embodiments, the etching utilized to form the voids **68** and **70** may adversely remove some of the charge-blocking material **46** from along the channel material pillars **36**. The dielectric material **72** may comprise a same composition as the charge-blocking material, and may be utilized to replace any of the charge-blocking material which may have been removed during formation of the voids **68** and **70**.

In some embodiments, the dielectric material **72** may be omitted.

Referring to FIG. **12**, conductive material **74** is formed within the opening **62**, and within the first and second void **68** and **70**. The conductive material **74** may be referred to as a second conductive material to distinguish it from the first conductive material **50**.

In the shown embodiment, the conductive material **74** comprises a first composition **76** and a second composition **78**. In some embodiments, the conductive material **74** may comprise the same compositions as the conductive material **50**. Accordingly, the composition **76** of conductive material **74** may be the same as the composition **52** of conductive material **50**; and may, for example, comprise, consist essentially of, or consist of metal nitride (e.g., titanium nitride). Also, the composition **78** of conductive material **74** may be the same as the composition **54** of conductive material **50**; and may, for example, comprise, consist essentially of, or consist of metal (e.g., tungsten). In some embodiments, the composition **76** of conductive material **74** may be referred to as a liner, and the composition **78** may be referred to as a core material formed along such liner.

The second conductive material **74** within the voids **68** and **70** may be considered to form additional gate levels **80** along the intervening regions **32**.

Referring to FIG. **13**, an opening **82** is formed through the second conductive material **74**, with such opening punching through the bottommost of the gate levels **30**. The opening **82** may be referred to as a second opening to distinguish it from the first opening **62** described above with reference to FIG. **9**.

The second opening **82** has the first side **63** and the opposing second side **65**. The second opening **82** isolates the gate levels **16** and **80** along the first side **63** from the gate levels **16** and **80** along the second side **65**.

Referring to FIG. **14**, the opening **82** is lined with a conductive material **84**. The conductive material **84** may comprise any suitable electrically conductive composition (s), such as, for example, one or more of various metals (e.g., titanium, tungsten, cobalt, nickel, platinum, ruthenium, etc.), metal-containing compositions (e.g., metal silicide, metal nitride, metal carbide, etc.), and/or conductively-doped semiconductor materials (e.g., conductively-doped silicon, conductively-doped germanium, etc.). In some embodiments, the conductive material **84** may comprise, consist essentially of, or consist of metal (e.g., tungsten). The conductive material **84** may be referred to as a third conductive material to distinguish it from the first and second conductive materials **50** and **74**.

Referring to FIG. **15**, the conductive material **84** is subjected to an anisotropic etch to form conductive structures **86** and **88** along sidewalls of the second opening **82**. The conductive structures **86** and **88** may be referred to as first and second conductive structures, respectively. The first conductive structure **86** couples the gate levels **30** and **80** along the first side **63** of the opening **82** into a first conductive structure corresponding to the select gate **64**; and

the second conductive structure **88** couples the gate levels **30** and **80** along the second side **65** of the opening **82** into a second conductive structure corresponding to the select gate **66**. The select gates **64** and **66** may be SGD's. An advantage of the SGD's **64** and **66** relative to conventional devices is that the SGD's **64** and **66** have long channel regions **90** due to the SGD's **64** and **66** comprising multiple conductive levels (**30/80**). In contrast, conventional architectures may utilize multiple short-channel SGD's in series, instead of the long-channel SGD's provided in the example embodiment architecture of FIG. **15**. The long-channel SGD's may be advantageous relative to conventional SGD's in that they may alleviate or prevent short-channel effects.

The dielectric material **72** is shown to remain along the top of construction **10** after the anisotropic etch of material **84**. In other embodiments, such dielectric material may be removed from over the top of the construction **10**. Also, one or both of the materials **56** and **60** may be removed from over the top of construction **10**.

In some embodiments, the channel material pillars **36** along the left side of the opening **82** are part of a first sub-block **100**, and the channel material pillars **36** along the right side of the opening **82** are part of a second sub-block **102**. The sub-blocks **100** and **102** may each correspond to a region coupled with a single sub-SGD line (such as one of the lines **342**, **344** and **346** of FIG. **3**). The SGD gates **64** and **66** may be together considered to be comprised by a select gate configuration **104**; with such configuration having a first conductive gate structure **64** associated with the first sub-block **100**, and having a second conductive gate structure **66** associated with the second sub-block.

In some embodiments, the stack **12** may be considered to comprise a first stack **106** comprising alternating insulative levels **14** and conductive levels **37**. The channel material pillars **36** extend through the first stack, and the memory cells **39** are along the channel material pillars within the first stack. An insulative level **108** is over the first stack **106**. The select gate configuration **104** is over the insulative level **108**.

In some embodiments, the select gate configuration **104** may be considered to comprise the first and second conductive gate structures **64** and **66** laterally spaced from one another by an intervening insulative region **110** along the cross-section of FIG. **15**. The first conductive structure **64** is along a first side **103** of the intervening insulative region **110**, and the second conductive structure **66** is along an opposing second side **105** of the intervening insulative region.

The intervening insulative region **110** is a void region at the processing stage of FIG. **15**, but may be filled with insulative material (as discussed below with reference to FIG. **16**).

The first conductive gate structure **64** has vertically-spaced conductive regions **112**, which are separated from one another by dielectric regions **114**. The first conductive structure **64** also has the vertically-extending conductive structure **86** which electrically couples the vertically-spaced conductive regions **112** with one another. Similarly, the second conductive gate structure **66** comprises the vertically-spaced conductive regions **116**, which are separated from one another by dielectric regions **118**; and has the vertically-extending conductive structure **88** coupling the vertically-spaced conductive regions **118** with one another.

In the illustrated embodiment of FIG. **15**, the intervening insulative region **110** includes a bottom portion which penetrates into the insulative level **108**. In other embodiments, the intervening insulative region **110** may extend to

an upper surface of the insulative level **108** rather than penetrating into the insulative level.

In some embodiments, the levels **30** and **32**, together with the dielectric material **72** between such levels, may be considered to be a second stack **118** over the first stack **106**; and specifically, over the insulative level **108**. The second stack **118** may be considered to comprise the conductive levels **30** and **32** as second conductive levels; and to comprise the dielectric material **72** as second insulative levels **120** which alternate with the second conductive levels.

The channel material pillars **36** extend through the first stack **106** and into the second stack **118**.

The opening **110** extends through the second stack **118**, but not entirely through the insulative level **108**. The opening **82** has the first side **63** and the second side **65** along the cross-section of FIG. **15**. The first conductive structure **86** may be considered to be a first conductive lining along the first side **63** of the opening **82**, and the second conductive structure **88** may be considered to be a second lining along the second side **65** of the opening **82**.

The first conductive lining **86** couples the conductive levels **30** and **32** along the first side **63** of the opening **82** to form a first select gate **64** for the first sub-block **100**; and the second conductive lining **88** couples the conductive levels **30** and **32** along the second side **65** of the opening **82** to form a second select gate **66** for the second sub-block **102**.

In the embodiment of FIGS. **5-15**, the first and second linings **86** and **88** are formed of the same material. In other embodiments, the first and second linings **86** and **88** may be formed of different materials relative to one another if such is desired in order to tailor the select gates **64** and **66** for specific purposes.

Referring to FIG. **16**, insulative material **190** is formed within opening **82** and over the conductive structures **86** and **88**. Also, conductive interconnects **92** are formed to extend through the insulative materials **56**, **60**, **72** and **190**, and to connect with the interconnects **38** along the tops of the channel material pillars **36**. The conductive interconnects **92** may couple the channel material pillars **36** with other circuitry (not shown in FIG. **16**).

The assemblies and structures discussed above may be utilized within integrated circuits (with the term "integrated circuit" meaning an electronic circuit supported by a semiconductor substrate); and may be incorporated into electronic systems. Such electronic systems may be used in, for example, memory modules, device drivers, power modules, communication modems, processor modules, and application-specific modules, and may include multilayer, multi-chip modules. The electronic systems may be any of a broad range of systems, such as, for example, cameras, wireless devices, displays, chip sets, set top boxes, games, lighting, vehicles, clocks, televisions, cell phones, personal computers, automobiles, industrial control systems, aircraft, etc.

Unless specified otherwise, the various materials, substances, compositions, etc. described herein may be formed with any suitable methodologies, either now known or yet to be developed, including, for example, atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), etc.

The terms "dielectric" and "insulative" may be utilized to describe materials having insulative electrical properties. The terms are considered synonymous in this disclosure. The utilization of the term "dielectric" in some instances, and the term "insulative" (or "electrically insulative") in other instances, may be to provide language variation within this disclosure to simplify antecedent basis within the claims

that follow, and is not utilized to indicate any significant chemical or electrical differences.

The particular orientation of the various embodiments in the drawings is for illustrative purposes only, and the embodiments may be rotated relative to the shown orientations in some applications. The descriptions provided herein, and the claims that follow, pertain to any structures that have the described relationships between various features, regardless of whether the structures are in the particular orientation of the drawings, or are rotated relative to such orientation.

The cross-sectional views of the accompanying illustrations only show features within the planes of the cross-sections, and do not show materials behind the planes of the cross-sections, unless indicated otherwise, in order to simplify the drawings.

When a structure is referred to above as being “on” or “against” another structure, it can be directly on the other structure or intervening structures may also be present. In contrast, when a structure is referred to as being “directly on” or “directly against” another structure, there are no intervening structures present.

Structures (e.g., layers, materials, etc.) may be referred to as “extending vertically” to indicate that the structures generally extend upwardly from an underlying base (e.g., substrate). The vertically-extending structures may extend substantially orthogonally relative to an upper surface of the base, or not.

A memory block may be understood as a smallest unit of a NAND memory array which can be erased. A sub-block may be understood as a portion of a block, and in some embodiments may be a portion tied to a single sub-SGD line (such as one of the lines 342, 344 and 346 of FIG. 3).

Some embodiments include an assembly having a stack of alternating dielectric levels and conductive levels. Channel material pillars extend through the stack. Some of the channel material pillars are associated with a first sub-block and others of the channel material pillars are associated with a second sub-block. Memory cells are along the channel material pillars. An insulative level is over the stack. A select gate configuration is over the insulative level. The select gate configuration includes a first conductive gate structure associated with the first sub-block, and includes a second conductive gate structure associated with the second sub-block. The first and second conductive gate structures are laterally spaced from one another by an intervening insulative region along a cross-section. The first conductive gate structure is along a first side of the intervening insulative region, and the second conductive gate structure is along a second side of the intervening insulative region. The second side is in opposing relation to the first side along the cross-section. The first and second conductive gate structures have vertically-spaced conductive regions separated from one another by dielectric regions, and have vertically-extending conductive structures which electrically couple the vertically-spaced conductive regions to one another.

Some embodiments include an assembly having a first stack comprising alternating first dielectric levels and first conductive levels. An insulative level is over the first stack. A select gate configuration is over the insulative level. The select gate configuration includes a second stack comprising alternating second dielectric levels and second conductive levels. Channel material pillars extend through the first stack and into the second stack. Some of the channel material pillars are associated with a first sub-block, and others of the channel material pillars being associated with a second sub-block. Memory cells are along the channel material pillars. An opening extends through the second stack, but not

entirely through the insulative level. The opening, along a cross-section, has a first side and an opposing second side. A first conductive lining is along the first side of the opening and couples the second conductive levels along the first side of the opening into a first select gate for the first sub-block. A second conductive lining is along the second side of the opening and couples the second conductive levels along the second side of the opening into a second select gate for the second sub-block.

Some embodiments include a method of forming an assembly. A stack of alternating first and second levels is formed. The first levels comprise dielectric material and the second levels comprise sacrificial material. Two of the uppermost second levels are gate levels, and a first level between said two of the uppermost second levels is an intervening first level. At least some of the second levels beneath the gate levels are wordline levels. Channel material pillars are formed to extend through the wordline levels, through at least one of the gate levels, and through the intervening first level. Slits are formed to extend through the stack. The sacrificial material of the second levels is replaced with first conductive material, and then the slits are filled with insulative material. A first opening is formed to extend through the first conductive material of at least one of the gate levels, and to extend through the intervening first level. The first opening defines a first gate side along a first side of the first opening, and defines a second gate side along a second side of the first opening. The second side of the first opening is in opposing relation to the first side of the first opening along a cross-section. The dielectric material is removed from the intervening first level to form a first void along the first side of the first opening, and to form a second void along the second side of the first opening. Second conductive material is formed within the first opening and within the first and second voids. The second conductive material within the first void is an additional gate level along the first gate side, and the second conductive material within the second void is an additional gate level along the second gate side. Some of the second conductive material is removed to form a second opening which isolates the gate levels of the first gate side from the gate levels of the second gate side. A first conductive structure is formed within the second opening and along the first gate side. The first conductive structure couples the gate levels along the first gate side with one another to incorporate the gate levels along the first gate side into a first conductive gate. A second conductive structure is formed within the second opening and along the second gate side. The second conductive structure couples the gate levels along the second gate side with one another to incorporate the gate levels along the second gate side into a second conductive gate.

In compliance with the statute, the subject matter disclosed herein has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the claims are not limited to the specific features shown and described, since the means herein disclosed comprise example embodiments. The claims are thus to be afforded full scope as literally worded, and to be appropriately interpreted in accordance with the doctrine of equivalents.

I claim:

1. An assembly, comprising:

a stack comprising alternating dielectric levels and conductive levels;

a channel material pillar extending through the stack; memory cells along the channel material pillar;

an insulative level over the stack;

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a select gate configuration over the insulative level, the select gate configuration comprising a conductive liner surrounding a conductive core; and
 a conductive interconnect extending through an uppermost surface of the channel material pillar.

2. The assembly of claim 1 wherein the conductive interconnect extends only partially into the channel material pillar.

3. The assembly of claim 1 wherein the conductive interconnect couples the channel material pillar to other circuitry.

4. The assembly of claim 1 further comprising vertically-extending conductive structures which electrically couple the conductive levels.

5. The assembly of claim 4 wherein the vertically-extending conductive structures extend through the insulative level over the stack.

6. The assembly of claim 1 further comprising an insulative layer extending over the alternating dielectric levels and conductive levels.

7. The assembly of claim 6 wherein the conductive interconnect extends into insulative layer.

8. The assembly of claim 1 further comprising an insulative layer extending over channel material pillars.

9. The assembly of claim 1 wherein the select gate configuration comprises a first conductive gate structure and a second conductive gate structure that comprises a different structural configuration relative to the first conductive gate structure.

10. The assembly of claim 1 further comprising an insulative liner against the insulative level.

11. The assembly of claim 1 further comprising an insulative liner over the insulative level and against an uppermost surface of the channel material pillar.

12. The assembly of claim 1 further comprising an insulative liner over the select gate configuration and the conductive interconnect extends through the insulative liner.

13. The assembly of claim 12 further comprising insulative material over the insulative liner and the conductive interconnect extends into the insulative material.

14. The assembly of claim 1 further comprising protective capping material over the select gate configuration and the conductive interconnect extends into the protective capping material.

15. The assembly of claim 1 further comprising dielectric material over the select gate configuration and the conductive interconnect extends into the dielectric material.

16. The assembly of claim 1 further comprising a level of conductive material in between the at least two of the conductive levels.

17. The assembly of claim 16 wherein the level of the conductive material is differently configured than either of the at least two of the conductive levels.

18. The assembly of claim 1 further comprising a gate level between the at least two of the conductive levels.

19. The assembly of claim 1 further comprising insulative material and conductive material between the at least two of the conductive levels.

20. An assembly, comprising:
 a stack comprising alternating dielectric levels and conductive levels;
 a channel material pillar extending through the stack; memory cells along the channel material pillar;
 an insulative level over the stack;
 a select gate configuration over the insulative level;
 a conductive interconnect extending through an uppermost surface of the channel material pillar;

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wherein the select gate configuration comprises a first conductive gate structure and a second conductive gate structure laterally spaced from one another by an intervening insulative region;
 wherein the first and second conductive gate structures have vertically-spaced conductive regions separated from one another by dielectric regions; and
 vertically-extending conductive structures which electrically couple the vertically-spaced conductive regions to one another.

21. The assembly of claim 20 wherein the vertically-extending conductive structures extend through the intervening insulative region.

22. An assembly, comprising:
 a stack comprising alternating dielectric levels and conductive levels;
 a channel material pillar extending through the stack; memory cells along the channel material pillar;
 an insulative level over the stack;
 a select gate configuration over the insulative level;
 a conductive interconnect extending through an uppermost surface of the channel material pillar;
 wherein the select gate configuration comprises a first conductive gate structure and a second conductive gate structure laterally spaced from one another by an intervening insulative region;
 an insulative layer extending through the intervening insulative region and extending over the alternating dielectric levels and conductive levels; and
 wherein the conductive interconnect extends into insulative layer.

23. An assembly, comprising:
 a stack comprising alternating dielectric levels and conductive levels;
 a channel material pillar extending through the stack; memory cells along the channel material pillar;
 an insulative level over the stack;
 a select gate configuration over the insulative level;
 a conductive interconnect extending through an uppermost surface of the channel material pillar; and
 an insulative liner against the insulative level.

24. An assembly, comprising:
 a stack comprising alternating dielectric levels and conductive levels;
 a channel material pillar extending through the stack; memory cells along the channel material pillar;
 an insulative level over the stack;
 a select gate configuration over the insulative level;
 a conductive interconnect extending through an uppermost surface of the channel material pillar; and
 an insulative liner over the select gate configuration and the conductive interconnect extends through the insulative liner.

25. An assembly, comprising:
 a stack comprising alternating dielectric levels and conductive levels;
 a channel material pillar extending through the stack; memory cells along the channel material pillar;
 an insulative level over the stack;
 a select gate configuration over the insulative level;
 a conductive interconnect extending through an uppermost surface of the channel material pillar; and
 an insulative liner over the select gate configuration and the conductive interconnect extends through the insulative liner.

26. The assembly of claim 25 further comprising insulative material over the insulative liner and the conductive interconnect extends into the insulative material.

27. An assembly, comprising:
a stack comprising alternating dielectric levels and conductive levels;
a channel material pillar extending through the stack;
memory cells along the channel material pillar; 5
an insulative level over the stack;
a select gate configuration over the insulative level;
a conductive interconnect extending through an uppermost surface of the channel material pillar; and
protective capping material over the select gate configuration and the conductive interconnect extends into the protective capping material. 10
28. An assembly, comprising:
a stack comprising alternating dielectric levels and conductive levels; 15
a channel material pillar extending through the stack;
memory cells along the channel material pillar;
an insulative level over the stack;
a select gate configuration over the insulative level;
a conductive interconnect extending through an uppermost surface of the channel material pillar; and 20
dielectric material over the select gate configuration and the conductive interconnect extends into the dielectric material.

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